CLASSIFICATION PERFORMANCE METRICS

- Binary classification setting

$y \in \{0, 1\}$

1) <u>Accuracy</u> : % of samples correctly classified.

2) <u>Balanced accuracy (BA)</u>

$Acc_0 \Rightarrow$ % of negative samples that are correctly classified.

$Acc_1 \Rightarrow$ " " positive " " "

$$BA = \frac{Acc_0 + Acc_1}{2}$$

<u>Dataset</u> 95% 1s, 5% 0s $\rightarrow \hat{y} = 1$ always

Accuracy = 95%

$$BA = \frac{0 + 100}{2} = 50\%$$

3) <u>Confusion Matrix</u>

True label	0	1
0	35 TN	10 FP
1	5 FN	25 TP

- True Positives (TP): Positive samples predicted well.
- False Positives (FP): Neg. samples predicted as positive.
- True Neg (TN): Neg. samples predicted as Neg.

Predicted label

Negatives Positives

- False Neg (FN): Pos. samples predicted as neg.

4) Precision, Recall, & F1-score

$$\text{Precision (P)} = \frac{TP}{TP + FP} = \frac{25}{35}$$

→ % of samples pred. as positive that are indeed positive.

$$\text{Recall (R)} = \frac{TP}{TP + FN} = \frac{25}{30}$$

or Sensitivity (S)

→ From the + samples, % that is correctly predicted.

$$\text{F1-score} = 2\,\frac{P \cdot R}{P + R}$$ ⇒ Harmonic mean of Prec. & Recall.

5) Precision-Recall curve

With soft classifiers, in particular Log. Reg,

$$h_\theta(x) = p(y=1 \mid x; \theta)$$

$= 1$ if > 0.5 = threshold

$$\hat{y} = \begin{cases} 1, & \text{if } h_\theta(x) \geq 0.5 \\ 0, & \text{oth.} \end{cases}$$ = threshold

Depending on the application, it may be desirable to increase or decrease this threshold.

	0	1
0	35 + 9 30	~~10~~ 1 +5
1	5 + 5 0	~~25~~ 20 +5

thr = 0.5

thr = 0.9 $> 0.9 \rightarrow \hat{y} = 1$

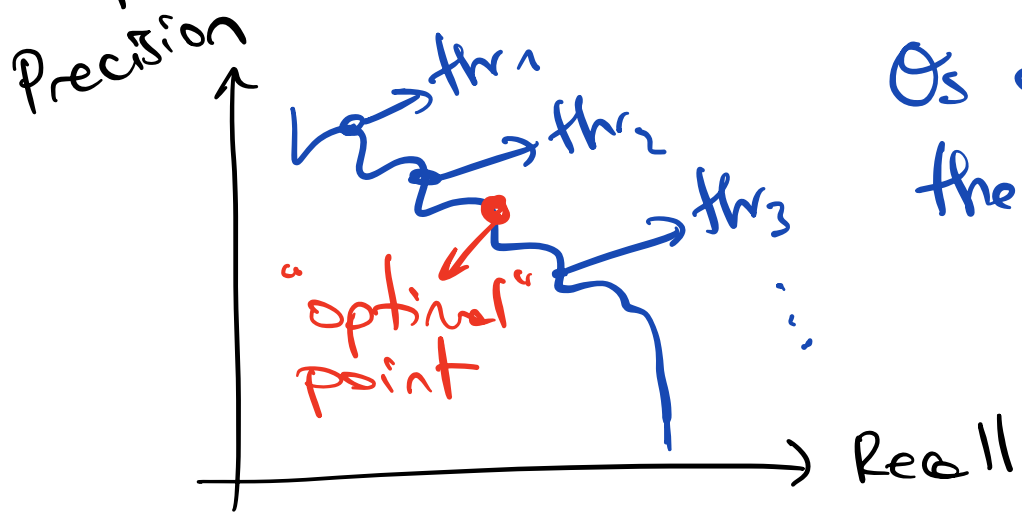
↳ increase the Precision

↳ it may result in a decrease of RECALL.

thr = 0.1 $< 0.1 \rightarrow \hat{y} = 0$

By modifying the threshold, you can increase or decrease the prec. & recall.

For soft classifiers, you can plot a prec.-recall curve by modifying the thr.



0s are always the same!!

6) ROC curve (Receiver Operating Characteristic curve)

$$= \frac{FP}{FP + TN}$$

True \ Predicted	0	1
0	TN	FP
1	FN	TP

→ FPR ↓↓

→ Recall ↑↑

AUROC → Area under the ROC

or **AUC**

$AUC_{random} = 0.5$

$AUC_{perf} = 1$

$AUC \in (0.5, 1)$ Typ. values 0.7, 0.8, 0.9...

The AUC gives you a sense of whether the model can "sort" the samples based on the probability.

$h_\theta(x)$	y		
0.99	1	0	1
0.9	1	0	1
0.87	1	1	1
0.85	1	1	1
0.76	1	1	1
0.6	1	1	1
0.58	1	1	0
0.55	1	1	0
0.48	1	0	1
0.3	0	0	1
0.21	0	0	0
0.10	0	0	0
0.05	0	1	0
0.02	0	1	0

AUC = 1

Filtering scheme:

Not predict samples whose $h_\theta(x) \in [0.4, 0.6]$ for example.